

Curriculum Vitae

July 22, 2026



Personal Information

Name

Thomas (Tom) T.P. Franken

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<https://www.linkedin.com/in/tomfranken311>

Education & Research Experience

Technical University Eindhoven, PhD Trajectory

10/2021-06/2026

→ *Subject*: The Autonomous Data Language: data-autonomous programming with a focus on structure, simplicity, parallelism and verification. Design, Adaptation and the creation of Verification Methods.

→ *Supervisors*: dr. ir. Thomas Neele (supervisor), prof. dr. ir. Jan Friso Groote (promotor)

Technical University Eindhoven, Master Computer Science

09/2018-08/2021

→ *Final Project*: **Set System Oracles for Subtrajectory Clustering**

* *Subject*: How to create a data structure to function as a set system oracle for use in solving Subtrajectory Covering with Greedy Set Cover.

* *Supervisors*: dr. Kevin Buchin (TU Eindhoven), prof. dr. Anne Driemel (University Bonn).

→ *Internship*:

* *Subject*: Formally translating Hierarchical Equation Systems into Parameterized Boolean Equation Systems and the use of CHC solving on both HES and its equivalent PBES.

* *Supervisor*: Tim Willemse (TU Eindhoven)

Technical University Eindhoven, Bachelor Computer Science

09/2015-07/2018

→ Combined Majors Software Science & Web Science

Teaching Experience

Practical Tutor Computer Systems

2022-2024

Leading practical sessions and the grading therein, designing exam questions, efforts in further developing the course.

Student Assistant Logic & Set Theory

09/2017-10/2017 & 09/2020-11/2020

Grading weekly tests and assisting at tutor sessions.

Student Assistant Algorithms

11/2018-01/2019 & 09/2019-11/2019

Grading weekly homeworks and assisting at tutorial sessions.

Student Assistant Computer Systems

02/2018-04/2018

Grading during tutorial sessions, as well as chat monitoring and answering questions during lectures.

Other Professional Experiences

- Treasurer of E.P.A. Nexus 04/2024-03/2025
Bookkeeping and budgeting as a treasurer, also involved in activity planning.
- Member of the TU/e Mathematics and Computer Science Department PhD Council 10/2022-05/2025
Discussing departmental policies and problems and how they relate to PhDs.
- Chairman of the Institute for Programming research and Algorithmics PhD Council 06/2022-08/2025
Point of Contact for the IPA director, organising the social event on the IPA Fall Days.
- Secretary of Da Vinci, Archery Association Eindhoven 11/2016-10/2017, 10/2020-09/2021
Answering mails, doing member administration, planning activities and making minutes of meetings.
- Member of Student Team T.E.S.T. (TU/e Sensing Team) 11/2017-09/2018
A student team project to create a biosensor for the SensUs competition.

Conferences and Reviews

Attended

- SEFM 2023 (*volunteer*)
- ICTAC 2023 (*accepted paper*)
- FORTE 2024 (*accepted paper*)

Reviewed for

- SETTA 2022 & 2023 (*subreviewer*)
- FMICS 2022 (*subreviewer*)
- iFM 2023 (*subreviewer*)

Skills and Interests

Languages

- Dutch, *native language*
- English, *fluent*
- German, *basic*
- Italian, *basic*

Research Interests

- Verification (of Concurrent Programs)
- Logics for Program Verification
- Semantics and Language Design
- Expressiveness and Programmability
- Graphs & Graph Games

Publications

- [1] Tom T. P. Franken, Thomas Neele, and Jan Friso Groote. “An Autonomous Data Language”. In: *ICTAC 2023 Proc.* Vol. 14446. LNCS. Springer Nature Switzerland, 2023, pp. 158–177. DOI: 10.1007/978-3-031-47963-2_11.
- [2] Tom T. P. Franken and Thomas Neele. “AuDaLa is Turing Complete”. In: *FORTE 2024 Proc.* Vol. 14678. LNCS. Springer Nature Switzerland, 2024, pp. 221–229. ISBN: 978-3-031-62645-6. DOI: 10.1007/978-3-031-62645-6_12.
- [3] Tom T. P. Franken, Thomas Neele, and Jan Friso Groote. “The Autonomous Data Language – Concepts, design and formal verification”. In: *Theoretical Computer Science* 1057 (2025), p. 115560. ISSN: 0304-3975. DOI: 10.1016/j.tcs.2025.115560.
- [4] Gijs P. Leemrijse, Tom T. P. Franken, and Thomas Neele. “Formalisation of a New Weak Semantics for AuDaLa”. In: *ATVA 2024 Proc.* Cham: Springer Nature Switzerland, 2025, pp. 93–116. ISBN: 978-3-031-78750-8. DOI: 10.1007/978-3-031-78750-8_5.
- [5] Tom T. P. Franken and Thomas Neele. “Expressivity of AuDaLa: Turing Completeness and Possible Extensions”. In: *Logical Methods in Computer Science* Volume 22, Issue 2 (2026). ISSN: 1860-5974. DOI: 10.46298/lmcs-22(2:5)2026.